PMSCS 623P

Quiz – 1 Makeup

Class Roll: ______________________ **Name:** ______________________

1. **Suppose you have the following list of numbers to sort**? 2
   [15, 5, 4, 18, 12, 19, 14, 10, 8, 20] which list represents the partially sorted list after three complete passes/iterations of insertion sort?

   (A) [4, 5, 12, 15, 14, 10, 8, 18, 19, 20] (C) [15, 5, 4, 10, 12, 8, 14, 18, 19, 20]

   (B) [4, 5, 15, 18, 12, 19, 14, 10, 8, 20] (D) [15, 5, 4, 18, 12, 19, 14, 8, 10, 20]

2. Analyze the running time of the following algorithm and write it using ***O***() notation (you must show the calculation/derivation of your running time). 3

```
for(int i=1; i <= n; i=i+1){
    for(int j=n; j<n; j=j--){
        printf("PMSCS 616");
        for(k=n/2; k<n; k=k++){
            sum++;
        }
    }
}
```

3. We can measure the actual running time of the execution of an algorithm. (True / False) 1

4. Fill in the following blanks with either True or False: 4

f(n)	***g(n)***	***f = O(g(n))***	***f = Ω(g(n))***
$n^2 + 45n + 1$	$(n^2/100) + 2n\sqrt{n} + 1$		
$n + logn^2\, n$	$nlogn$		